

LUMING TANG

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EDUCATION

Cornell University, Department of Computer Science Jan.2019-Present
Ph.D. in Computer Science
Advisor: Prof. Bharath Hariharan

Tsinghua University, Department of Physics Sep.2014-Jun.2018
B.S. in Mathematics and Physics
Double Major in Economics

PUBLICATION

Luming Tang, Davis Wertheimer, Bharath Hariharan. “Revisiting Pose-Normalization for Fine-Grained Few-Shot Recognition”, in *Conference on Computer Vision and Pattern Recognition (CVPR 2020)*.

Luming Tang, Yexiang Xue, Di Chen, Carla P. Gomes. “Multi-Entity Dependence Learning with Rich Context via Conditional Variational Auto-encoder”, in *AAAI Conference on Artificial Intelligence (AAAI 2018)*.

Zhongdao Wang*, **Luming Tang***, Xihui Liu, Zhuliang Yao, Shuai Yi, Jing Shao, Junjie Yan, Shengjin Wang, Hongsheng Li, Xiaogang Wang. “Orientation Invariant Feature Embedding and Spatial Temporal Regularization for Vehicle Re-identification”(*equal contribution), in *International Conference on Computer Vision (ICCV 2017)*.

EXPERIENCE

Cornell University, Research Assistant May.2019-Present
Advisor: Bharath Hariharan

- Revisit pose normalization for fine grained few-shot recognition problem and show that with a minimal increase on model capacity, it can improve performance significantly for multiple different learning algorithms and network backbones.

Microsoft Research Asia, Research Intern Sep.2018-Dec.2018
Mentor: David Wipf

- Analyzed the regularization balance of Autoencoder-structured models in general and VAEs in particular. This leads to useful practical prescriptions and the first demonstration of high-quality, diverse generation results from Autoencoder-structured, non-adversarial training on high-resolution images.

Cornell University, Research Intern Jun.2017-Sep.2017
Advisor: Carla Gomes

- Proposed a variational auto-encoder based algorithm to model structured multi-entity distribution, achieving better performance on two real-world applications compared to previous state-of-the-art approximate inference based methods.

SenseTime, Research Intern Dec.2016-Jun.2017
Mentor: Shuai Yi

- Explored vehicle re-identification problem and proposed orientation invariant embedding together with spatial-temporal regularization. The model doubled the matching accuracy of previous methods on four public datasets.

- Researched video understanding problem and proposed the deep recurrent architecture, achieving significant performance in Kaggle Youtube-8M challenge. The first-author paper was accepted by CVPR Video Understanding Workshop.

Tsinghua University, Research Assistant

Sep.2016-Jun.2018

Advisor: Zhiyuan Liu

- Get involved in constructing an open-source framework OpenNRE for neural relation extraction with labmates. The code is available on THUNLP Github homepage, receiving over 1300 stars and 400 folks.

TEACHING

CS4787 Principles of Large-Scale Machine Learning, Teaching Assistant

Spring 2019

CS2110 OOP and Data Structures, Teaching Assistant

Summer 2019

CS6670 Graduate Computer Vision, Teaching Assistant

Fall 2019